

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Second Semester of M. Tech (CE) Examination

May 2015

CE714 Digital Image Processing

Date: 11.05.2015, Monday

Time: 01.30 p.m. To 04.30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q-1(a) Define following terms. [03]

(1) Convex hull (2) Radiance (3) Lossy Image compression

(b) Fill in the blanks. [02]

(1) Opening is _____ followed by erosion.

(2) _____ numbers of colors you can have in 3bit/pixel image.

(c) Answer the following questions. [06]

(1) What is data redundancy? Write equations to compression ratio and relative data redundancy for two data set having N_1 and N_2 information carrying units respectively.

(2) What is the minimum size of mask? Which mask is of size 2×2 ?

(3) List three basic types of gray level discontinuities in digital image.

Q-2 Answer the following questions. (Any two) [14]

(a) Explain thinning and thickening.

(b) Explain region splitting and merging with example.

(c) List different method of boundary description and explain any two of them.

(d) What is external and internal representation and when they are used? Also explain any two object representation methods.

Q-3 Answer the following questions. (Any two) [10]

(a) What is thresholding? Explain the concept using bimodal and multimodal histogram. Explain global thresholding.

(b) Write a short note on JPEG.

- (c) Explain Face recognition.

SECTION – II

Q-4(a) Define the following terms in context of image processing. [02]

- (1) Pixel (2) Histogram

(b) State that following statements are True or False. [03]

- (1) Power law transformation is also called gamma correction.
 (2) Laplacian filtering is using 1st order derivation.
 (3) Microwave images are used in Radar.

(c) Answer the following questions. [06]

- (1) How many bytes are required to store an image of 20x20 having 64 different intensity levels?
 (2) Explain sampling and quantization of gray level images.
 (3) Write equations to compute output intensity “s” for input intensity “r” using Image negation and Log transformation.

Q-5 Answer the following questions. (Any two) [14]

- (a) Find equalized histogram for given input histogram. Plot both input and equalized histogram.

Gray level	0	1	2	3	4	5	6	7
No. of pixels	190	250	210	160	80	60	30	20

- (b) What is image restoration? List and explain different spatial domain mean filters for image restoration.
 (c) Explain RGB, CMY, HSI color model and state where they are applicable.
 (d) Explain region filling and boundary extraction.

Q-6 Answer the following questions. (Any two) [10]

- (a) List down steps of frequency domain enhancement with block diagram.
 (b) Define bandreject and Bandpass filters. Write filter equation for Butterworth band reject filter of order “n” and Gaussian band reject filter. Also pen down the equation to derive Bandpass filter from Bandreject filter.
 (c) List three sharpening filters in frequency domain image enhancement. Explain Homomorphic filtering.
